

Exploring Knowledge Filtering for Retrieval-Augmented Question Answering

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Abstract—Retrieval-Augmented Generation significantly enhances the performance of Large Language Models on knowledge-intensive tasks by incorporating external knowledge. However, retrieval is not always beneficial, as the presence of harmful knowledge in retrieved documents can degrade generation quality. While existing methods attempt to mitigate this by either assessing the necessity of retrieval or by compressing retrieved documents, they struggle to reject entirely harmful documents selectively. To address this, we propose a novel two-stage framework. First, we establish an annotation strategy to label the harmfulness of a document based on its impact on downstream task performance. Subsequently, our Knowledge Filtering module utilizes this annotated data to learn to identify and eliminate harmful knowledge. For this module, we develop and evaluate four distinct strategies, ranging from methods that leverage LLM self-evaluation and lightweight proxy models to those based on preference alignment. Experiments conducted on three diverse question answering datasets demonstrate that our approach effectively purges harmful knowledge, leading to more reliable and accurate responses.

Index Terms—Retrieval-augmented generation, large language models, knowledge filtering, preference alignment.

I. INTRODUCTION

IN RECENT years, Large Language Models (LLMs) have achieved remarkable advances in the field of Natural Language Processing (NLP), particularly demonstrating powerful capabilities in text generation [2] and comprehension [3]. However, LLMs also exhibit certain inherent limitations, such as the propensity to generate factually inconsistent hallucinated content during generation [4], as well as their knowledge being constrained to the training data, rendering them unable to dynamically update their internal parametric knowledge [5]. To address these challenges, Retrieval-Augmented Generation (RAG) [6] has emerged as a promising approach. RAG enhances the generation process of large language models by retrieving relevant information from external knowledge bases (e.g.,

Wikipedia) and integrating it into the input context. This method has been shown to significantly improve the factual accuracy and relevance of generated responses, particularly in applications involving knowledge-intensive tasks such as Question Answering (QA) [7], [8].

Recent work reveals that RAG's retrieval process is not always beneficial [9]. When the LLM's parametric knowledge is sufficient for a query, external information can act as noise and degrade output quality. This has made determining when and how to utilize external knowledge a critical research focus, with current approaches falling into two main paradigms: 1) pre-retrieval adaptation [9], [10], seeks to determine the necessity of retrieval before a search is conducted, often by prompting the model for self-assessment or probing its internal states for uncertainty. However, the effectiveness of these methods is highly contingent on the retriever's performance and the completeness of the knowledge corpus, creating a risk that useful information may be erroneously omitted. 2) post-retrieval adaptation [11], [12], focuses on refining the retrieved documents before they are fed to the generator, typically through passage compression or extractive filtering. While these techniques can reduce noise, they were originally designed to condense information rather than discard it. Therefore, these methods fail to prevent the influence of documents whose content is pervasively irrelevant or harmful.

As shown in Fig. 1, this leads us to propose a critical question: *Can we devise a mechanism that integrates the advantages of the two existing paradigms?* Specifically, a filtering mechanism that operates by evaluating each retrieved document and selectively discarding it. Such a strategy would ensure all potentially relevant information is considered, thereby mitigating the inherent risk in pre-retrieval methods of prematurely omitting useful data. Simultaneously, it would possess the capability to reject an entire document if it is deemed harmful, overcoming the limitation of post-retrieval compression techniques that may retain harmful content. However, developing such a filter mechanism presents two significant challenges. On the one hand, there is the challenge of **how to annotate the harmfulness of a document?** The harmfulness of a document is not an absolute property but is co-determined by the specific query and the preference of LLMs being used, which makes developing a reliable annotation strategy a non-trivial task. On the other hand, there is the challenge of **how to effectively utilize this annotated data.** The goal is to devise a strategy that can accurately and efficiently learn this complex, preference-aware filtering capability. It needs to be lightweight enough not to introduce significant latency into the

Received 5 July 2025; revised 7 December 2025 and 15 January 2026; accepted 19 January 2026. Date of publication 29 January 2026; date of current version 9 February 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62376178, in part by the Jiangsu Key Laboratory of Language Computing (JSLCKeyLab under Grant 202500003, and in part by Priority Academic Program Development of Jiangsu Higher Education Institutions. An earlier version of this paper was presented in part at the Findings of the Association for Computational Linguistics: ACL 2025 [DOI: 10.18653/v1/2025.findings-acl.86]. The associate editor coordinating the review of this article and approving it for publication was Dr. Derek F. Wong. (Corresponding author: Zhongqing Wang.)

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Digital Object Identifier 10.1109/TASLPRO.2026.3658957

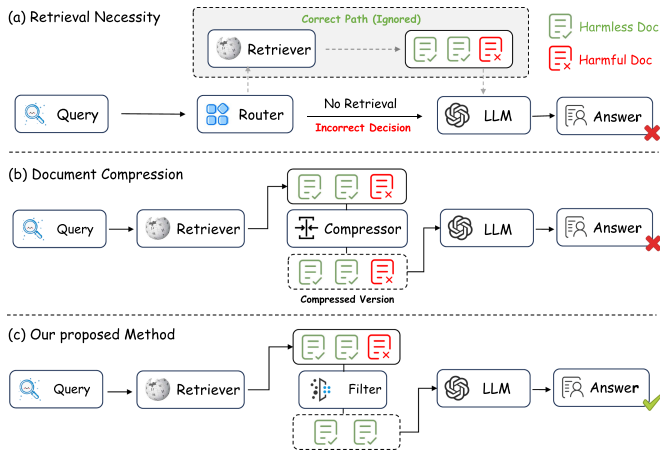


Fig. 1. Comparison of different strategies for handling retrieved documents in RAG. (a) Retrieval Necessity models may incorrectly decide not to retrieve, thereby missing crucial external knowledge. (b) Document Compression models shorten all retrieved documents but cannot completely eliminate a document that is entirely harmful. (c) Our Proposed Method evaluates each document individually, discarding harmful ones while retaining beneficial ones.

RAG pipeline, while also being powerful enough to comprehend the intricate relationship between the query and the retrieved documents.

In this study, we tackle these challenges by proposing a novel framework structured into two primary modules: **Label Collection** and **Knowledge Filtering**. To establish a reliable method for identifying harmful knowledge, our Label Collection module annotates query-document pairs based on the specific performance metric: whether the inclusion of the document improves or degrades the LLM’s response quality to the query. After collecting this annotated data, we introduce the Knowledge Filtering module, where we develop and investigate four distinct strategies designed to learn this filtering capability. These strategies range from leveraging the backbone LLM’s own reasoning and employing efficient proxy models to utilizing powerful preference alignment methods. Through this comprehensive framework, our work provides a robust solution for adaptively removing harmful knowledge from the retrieval context, thereby improving the overall performance of RAG systems. Extensive experiments conducted on three diverse Question Answering datasets validate the effectiveness of our proposed approach.

Difference from Conference Version. This manuscript represents a substantial extension of our preliminary work presented at ACL 2025 [1]. While our previous study introduced the concept of knowledge filtering for discriminative tasks (e.g., sentiment analysis) using the perplexity-based method, this work advances the paradigm to the more challenging domain of Generative Question Answering. This transition necessitates a fundamental overhaul of both the problem definition and methodology. Specifically, (1) **Task Evolution:** We shift from predicting discrete labels to generating free-form text, where “harmful knowledge” leads to complex hallucinations rather than simple classification errors. (2) **Methodological Innovation:** We propose a novel Preference-Aligned Scorer trained via Contrastive Learning and Direct Preference Optimization (DPO), replacing the previous ensemble approach.

Furthermore, we introduce a task-oriented annotation strategy based on F1-score changes, providing a more reliable signal than the previous perplexity-based metric. (3) **Comprehensive Evaluation:** We conduct extensive experiments on open-domain QA benchmarks (NQ, PopQA, AmbigNQ) with new generative baselines, demonstrating the robustness of our approach in knowledge-intensive scenarios.

II. RELATED WORK

A. Retrieval-Augmented Large Language Models

Large Language Models have achieved significant progress in various domains of Natural Language Processing [13]. However, they still face several inherent challenges, such as generating factual errors (i.e., “hallucinations”), their internal knowledge being static post-pretraining, and underperforming on tasks that require domain-specific knowledge [13], [14], [15]. To address these challenges, Retrieval-Augmented Generation has been widely adopted as a pivotal and effective approach. The foundational RAG framework, proposed by Lewis et al. [6], pioneered the integration of a pre-trained retriever [16] with a sequence-to-sequence generator [17]. This framework follows the *Retrieve-then-Read* paradigm: given user input, it retrieves relevant documents from a large-scale corpus (e.g., Wikipedia), and subsequently, these retrieved documents are fed into the generator along with the original input to produce more informative and accurate text. This approach has demonstrated exceptional performance on knowledge-intensive tasks, such as open-domain question answering [18], [19] and dialogue generation [20].

A crucial challenge in optimizing RAG systems is that unconditional retrieval is not always beneficial. When the LLM’s internal parametric knowledge is sufficient to address a query, retrieving external information can be redundant and may even introduce distracting noise, ultimately degrading the quality of the final generation [21]. Consequently, extensive research has focused on developing adaptive retrieval mechanisms, which dynamically control when and how external knowledge is utilized. These approaches can be broadly categorized into two main paradigms: pre-retrieval and post-retrieval adaptation: (1) **Pre-retrieval Adaptation:** This method focuses on determining the necessity of retrieval before executing a search. The primary goal is to create a gating mechanism that avoids retrieval altogether for queries that the LLM can handle independently. For instance, some methods leverage the LLM’s own self-assessment. The SKR framework [9] prompts the model to recognize its knowledge gaps and adaptively retrieve only when necessary. To mitigate the high cost of this self-check, others [22] employ a smaller slim proxy model to generate a preliminary answer, using it to infer the main LLM’s need for external knowledge. A different approach [10] bypasses textual generation and instead probes the model’s internal states. Probing-RAG, for example, inspects intermediate hidden representations to gauge model uncertainty, using it as a direct signal to trigger retrieval. (2) **Post-retrieval Adaptation:** This method focuses on refining the retrieved information after a search has been executed but before it is fed to the generator. The primary goal is to improve

the quality of response by compressing passages and filtering out irrelevant content. For instance, PRCA [23] inserts a pluggable module trained with reinforcement learning to compress passages for a black-box generator, optimizing directly for the final task reward. A more theoretically grounded approach [11] uses the information bottleneck principle to train a filter that explicitly maximizes information relevant to the answer while minimizing noise from the original retrieved text. A different approach [12] is extractive compression, which selects rather than rewrites information. EXIT, for example, performs context-aware sentence selection that can be parallelized, simultaneously improving QA accuracy and reducing latency.

B. Preference Alignment for Retrieval-Augmented Generation

The foundational goal of preference alignment is to steer the behavior of Large Language Models to better cohere with human values and instructions [24], [25]. A dominant paradigm for achieving this has been Reinforcement Learning from Human Feedback (RLHF), an approach that involves training an explicit reward model from preference data to guide the LLM's fine-tuning process [26]. However, the practical implementation of RLHF presents challenges, including the complexity and instability inherent in reinforcement learning and reward model training. In response to these difficulties, subsequent research [27], [28] has introduced more direct alignment methods, chief among them being Direct Preference Optimization (DPO) [29]. DPO reformulates the objective to prevent the need for a reward model, instead optimizing the LLM's policy directly on preference pairs.

A central challenge in advancing Retrieval-Augmented Generation systems is addressing the frequent misalignment between what the retriever provides and what the generator needs to craft a comprehensive and accurate answer. Consequently, extensive research [23], [28], [30], [31] has turned to preference alignment methods to harmonize these two critical components. Initial strategies focused on adapting individual components within the pipeline, often treating the core LLM as a fixed, black-box entity. For instance, some research [30] concentrates on the input to the retriever, proposing a Rewrite-Retrieve-Read framework where a small, trainable rewriter is optimized with reinforcement learning to better align the search query with the LLM reader's needs. An alternative approach [23] intervenes after retrieval, introducing a pluggable, reward-driven contextual adapter (PRCA) that refines the retrieved information before it reaches the generator. While these component-level adaptations offer improvements, more recent research has pursued more integrated optimizations. Shifting focus to the generator, PARAG [31] directly fine-tunes the LLM using a combination of Supervised Fine-tuning (SFT) and Direct Preference Optimization on multi-perspective preference data to enhance qualities like informativeness and citation accuracy. Taking this holistic view a step further, the DPA-RAG framework [28] proposes a comprehensive dual-alignment strategy: it simultaneously aligns a reranker to refine document selection and fine-tunes the LLM to better process the provided knowledge, creating a versatile

system that can adapt to the diverse preferences of various LLMs.

III. METHOD

To address the challenge of filtering harmful knowledge from retrieved documents, we propose a novel framework that operates in two main stages, as illustrated in Fig. 2. First, we devise a Label Collection module to create a dataset where the harmfulness of the document is explicitly defined by its impact on the LLM's performance on the downstream task. Second, we introduce a Knowledge Filtering module that utilizes annotated data to train a filter, for which we propose and investigate four distinct strategies spanning from LLM-based methods to preference alignment. Below, we will first introduce the task formulation and the core components of our system, the Retriever and Generator (Section III-A). We will then detail our method for Label Collection (Section III-B) and the proposed Knowledge Filtering strategies (Section III-C), and finally, present the complete inference pipeline of our framework (Section III-D).

A. Task Formulation & Framework

Given a user query x and a large external knowledge corpus $\mathcal{D} = \{d_1, d_2, \dots, d_N\}$, the objective of our task is to generate an accurate and reliable answer $\hat{y}$.

The process begins with a retrieval step. A retriever $\mathcal{R}$ takes the query x as input and fetches a set of k potentially relevant documents from the corpus, which we denote as $\mathcal{D}_{\text{ret}} = \mathcal{R}(x) = \{d_1, d_2, \dots, d_k\}$, where $\mathcal{D}_{\text{ret}} \subset \mathcal{D}$. In our work, this is accomplished using the BM25 algorithm, a sparse retrieval method that scores documents based on the statistical relevance of the query terms they contain. Unlike a standard RAG pipeline which would proceed directly with $\mathcal{D}_{\text{ret}}$, our framework introduces a critical Knowledge Filter module $\mathcal{F}$. The purpose of this filter is to identify and discard documents from $\mathcal{D}_{\text{ret}}$ that are deemed harmful for the LLM to answer the query x . The filter $\mathcal{F}$ takes the set of retrieved documents and produces a refined, filtered subset $\mathcal{D}_{\text{filt}}$:

$$\mathcal{D}_{\text{filt}} = \mathcal{F}(\mathcal{D}_{\text{ret}}) \quad (1)$$

where $\mathcal{D}_{\text{filt}} \subset \mathcal{D}_{\text{ret}}$. In cases where all retrieved documents are identified as harmful, the resulting set $\mathcal{D}_{\text{filt}}$ becomes empty. Finally, the filtered set of documents $\mathcal{D}_{\text{filt}}$ is concatenated with the original query x and fed into a large language model $\mathcal{M}$ to generate the final answer $\hat{y}$:

$$\hat{y} = \mathcal{M}(x, \mathcal{D}_{\text{filt}}) \quad (2)$$

If the filter discards all documents ($\mathcal{D}_{\text{filt}} = \emptyset$), the process defaults to standard generation using only the LLM's internal parametric knowledge.

In the following, we introduce the two fundamental components of the RAG pipeline: the retriever and the generator.

Retriever: For the retrieval component $\mathcal{R}$, we employ BM25 [32], a classical and highly effective sparse retrieval algorithm. BM25 ranks documents based on their lexical overlap with the query. The relevance score for a document d given a

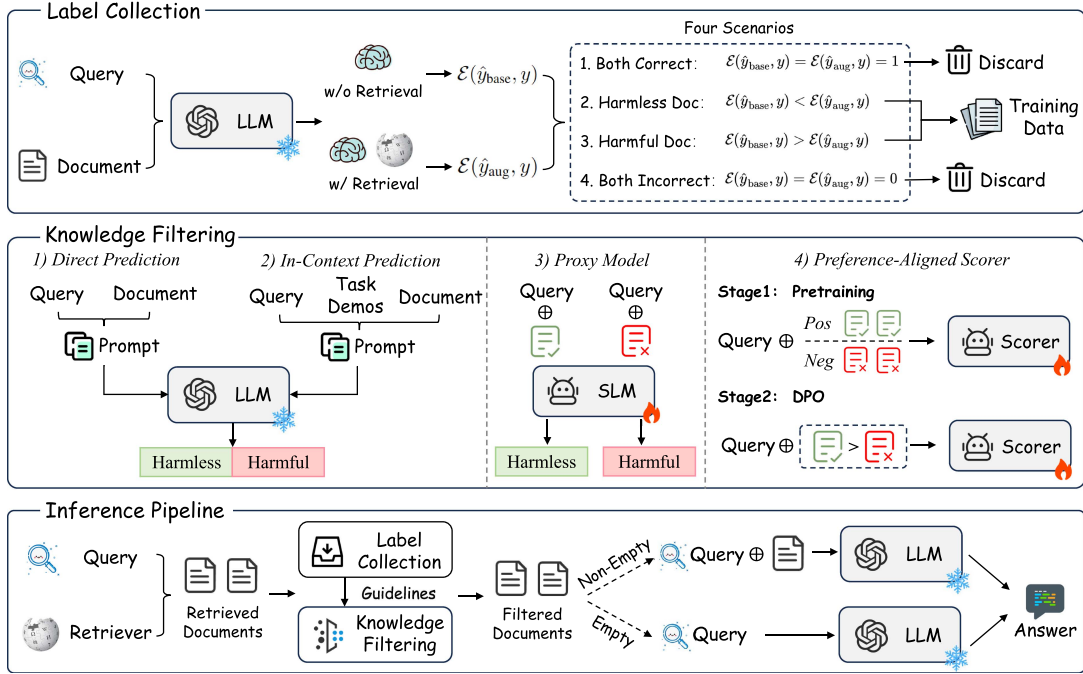


Fig. 2. An overview of our proposed framework for harmful knowledge filtering in RAG.

query x , which is composed of terms t_i , is calculated as:

$$\text{score}(d, x) = \sum_{t_i \in x} \text{IDF}(t_i) \cdot \frac{f(t_i, d) \cdot (k_1 + 1)}{f(t_i, d) + k_1 \cdot \left(1 - b + b \cdot \frac{|d|}{\text{avdgl}}\right)} \quad (3)$$

where $\text{IDF}(t_i)$ is the inverse document frequency of term t_i , $f(t_i, d)$ is the term's frequency in document d , $|d|$ is the length of the document, and avdgl is the average document length in the corpus. We use the standard parameters $k_1 = 1.5$ and $b = 0.75$. Based on this score, we select the top- k documents to form the initial retrieved set $\mathcal{D}_{\text{ret}}$.

Generator: We employ LLaMA [33] and Qwen [34] as the backbone of our framework. After the retrieval and subsequent filtering stages, the refined knowledge set $\mathcal{D}_{\text{filt}}$ is concatenated with the original query x to construct a single prompt. This prompt is then fed into the language model to generate the final answer $\hat{y}$:

$$\hat{y} = \mathcal{M}(\text{Prompt}(x, \mathcal{D}_{\text{filt}})) \quad (4)$$

The generator $\mathcal{M}$ operates without task-specific fine-tuning, ensuring an unbiased evaluation that isolates the direct impact of our knowledge filtering stage. This approach allows us to directly attribute any improvements in generated answers to the effectiveness of our filtering strategy in refining the input context for the LLM.

B. Label Collection

A key prerequisite for training our knowledge filter is a dataset with reliable labels indicating whether a document is harmful or harmless for a given query. To this end, we propose an annotation strategy that defines a document's utility based on its direct

impact on the generator's performance. The process is applied to each query-document pair (x, d) from the retrieved set $\mathcal{D}_{\text{ret}}$, using a ground-truth answer y from the training data.

Specifically, for a given pair $(x, d) \in \mathcal{D}_{\text{ret}}$, we first prompt the LLM to generate a base answer $\hat{y}_{\text{base}} = \mathcal{M}(x)$, which relies solely on the model's internal, parametric knowledge. We then generate an augmented answer $\hat{y}_{\text{aug}} = \mathcal{M}(x, d)$, using the context provided by the retrieved document. We evaluate both answers against the ground-truth label y using the evaluation metric $\mathcal{E}$, which in our case is the F1 score. Inspired by work on constructing preference data [9], we categorize the outcome based on the performance change to derive a clear and unambiguous signal of the document's effect. The label for the document d with respect to the query x is determined as follows:

$$\text{Label}(d, x) = \begin{cases} \text{Harmless} & \text{if } \mathcal{E}(\hat{y}_{\text{base}}, y) < \mathcal{E}(\hat{y}_{\text{aug}}, y) \\ \text{Harmful}, & \text{if } \mathcal{E}(\hat{y}_{\text{base}}, y) > \mathcal{E}(\hat{y}_{\text{aug}}, y) \\ \text{Discard}, & \text{otherwise} \end{cases} \quad (5)$$

We explicitly discard pairs where the performance is unchanged ($\mathcal{E}(\hat{y}_{\text{base}}, y) = \mathcal{E}(\hat{y}_{\text{aug}}, y)$). This includes cases where both answers are correct or both are incorrect, as these scenarios provide an ambiguous or noisy signal regarding the document's specific contribution. By focusing only on instances where the document causes a clear performance swing, we construct a high-quality training dataset $\mathcal{D}_{\text{kf}}$, containing query-document pairs reliably labeled as either "Harmful" or "Harmless."

C. Knowledge Filtering

With the annotated dataset $\mathcal{D}_{\text{kf}}$ constructed, we can now train a model to perform the filtering task. We propose and investigate four distinct strategies of increasing complexity to accomplish this: *Direct Prediction*, *In-Context Prediction*, *Proxy Model*,

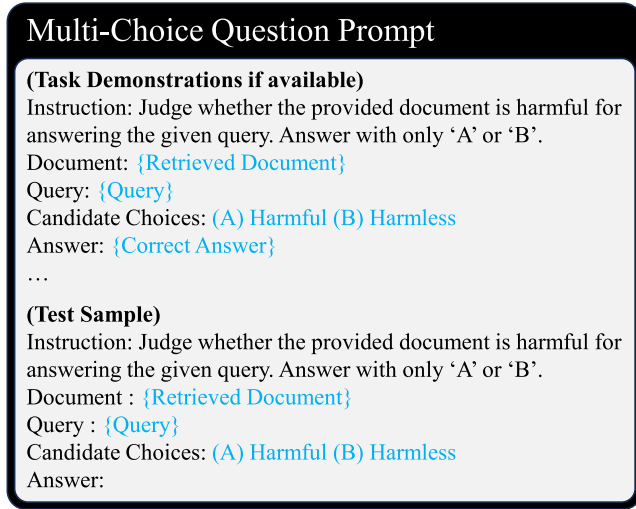


Fig. 3. Illustration of the Multiple-Choice Question prompt template.

and *Preference-Aligned Scorer*. A detailed description of each strategy is provided below.

1) *Direct Prediction*: The most straightforward approach is to leverage the LLM itself to directly assess the harmfulness of a given document in a zero-shot setting. The core assumption is that the LLM, given its vast training, possesses the intrinsic reasoning ability to perform this evaluation. To elicit a structured judgment, we frame this task as a Multiple-Choice Question (MCQ) [35]. As shown in Fig. 3, a carefully designed prompt instructs the model to analyze the provided query and retrieved document, and then choose between predefined options corresponding to our “Harmful” and “Harmless” labels.

We utilize the MCQ format because LLMs are generally proficient at this type of constrained selection task, often yielding more consistent results than open-ended evaluations. While this strategy is simple and requires no training on our labeled data, its effectiveness is inherently limited as it relies solely on the model’s general, pre-existing knowledge without any task-specific demonstrations to guide its reasoning.

2) *In-Context Prediction*: To improve upon the zero-shot baseline, our second strategy incorporates task-specific demonstrations into the prompt. Building on the principle of in-context learning (ICL) [36], which has been shown to significantly enhance LLM performance without fine-tuning, we augment the prompt with a few examples. We randomly sample several labeled query-document pairs from our constructed dataset $\mathcal{D}_{kf}$ and format them as demonstrations. These few-shot exemplars, each consisting of a query, a document, and its corresponding “Harmful” or “Harmless” label, are prepended to the prompt before the target pair that needs evaluation. The LLM is then expected to leverage these examples to better understand the task and make a more informed prediction.

However, both this strategy and Direct Prediction, which rely on the LLM’s direct inference capabilities, exhibit significant limitations. First, their performance is highly sensitive to the quality and selection of the demonstrations provided in the prompt [37]. Second, they are constrained by the finite context

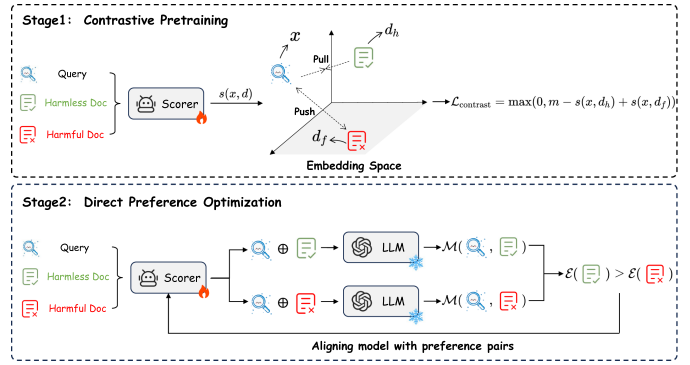


Fig. 4. The two-stage training process for our Preference-Aligned Scorer.

window of the LLM, which makes it impractical to leverage a large number of training examples and thus limits the potential for robust generalization.

3) *Proxy Model*: To overcome the context length and sensitivity limitations of direct LLM inference, our third strategy involves training a dedicated, lightweight proxy model. This approach allows us to leverage the entire labeled dataset $\mathcal{D}_{kf}$ to create a specialized and efficient filter. For this purpose, we employ the DeBERTa-v3-base [38] model and reformulate the filtering task as a sequence classification problem. Specifically, for each query-document pair (x, d) , we construct a single input sequence by concatenating them, separated by a special $[SEP]$ token. The DeBERTa model, with a classification head on top, is then fine-tuned on $\mathcal{D}_{kf}$ to predict one of two labels: “Harmful” or “Harmless”. The model, which we denote as $\mathcal{M}_{\text{proxy}}$, is trained to minimize the cross-entropy loss, formally defined as:

$$\hat{y}_{\text{label}} = \mathcal{M}_{\text{proxy}}([CLS] \oplus x \oplus [SEP] \oplus d) \quad (6)$$

where $\oplus$ denotes concatenation and $\hat{y}_{\text{label}}$ is the predicted label. By offloading the judgment to a smaller, fine-tuned discriminative model, this strategy significantly reduces the computational overhead and inference latency compared to using the large generator LLM, while creating a filter explicitly optimized for our task.

4) *Preference-Aligned Scorer*: As our final strategy, we reframe the filtering task as a preference learning problem. The core idea is that our labeled data, “Harmful” and “Harmless,” implicitly defines the LLM’s preference: for a given query, it prefers a harmless document over a harmful one. To capture this, we propose training a dedicated Preference-Aligned Scorer $\mathcal{M}_{\text{score}}$, built upon an XLM-Roberta [39] backbone, as shown in Fig. 4. The model takes a query and a document as input and outputs a single scalar score, $s = \mathcal{M}_{\text{score}}(x, d)$, indicating the document’s utility. The training of our scorer model proceeds in two distinct stages:

Stage 1. Contrastive Pre-training: To endow the model with a foundational understanding of the semantic differences between the two classes, we first conduct contrastive learning on the dataset $\mathcal{D}_{kf}$. For each query x in our data, we form triplets of (x, d_h, d_f) , where d_h is a “Harmless” document and d_f is a “Harmful” one. The model is trained using a margin-based

contrastive loss, which aims to push the score of the harmless document higher than that of the harmful one by at least a margin m :

$$\mathcal{L}_{\text{contrast}} = \max(0, m - \mathcal{M}_{\text{score}}(x, d_h) + \mathcal{M}_{\text{score}}(x, d_f)) \quad (7)$$

This initial stage helps to structure the embedding space and provides a strong starting point for the subsequent fine-tuning.

Stage 2. Direct Preference Optimization: After the contrastive pre-training, we further refine the scorer using Direct Preference Optimization. We translate our labeled data into a preference format, where for each query, the harmless document is the chosen response (d_w) and the harmful document is the rejected response (d_l). The DPO training objective is to minimize the following loss, defined as the expectation over all preference pairs in our dataset $\mathcal{D}_{\text{kf}}$:

$$\mathcal{L}_{\text{DPO}} = -\mathbb{E}_{(x, d_w, d_l) \sim \mathcal{D}_{\text{kf}}} [\log \sigma(\mathcal{M}_{\text{score}}(x, d_w) - \mathcal{M}_{\text{score}}(x, d_l))] \quad (8)$$

where σ is the sigmoid function. This directly fine-tunes the model to align with the preference pairs, producing a more accurate scoring function.

During inference, each retrieved document $d_i \in \mathcal{D}_{\text{ret}}$ is scored individually. To maintain consistency with previous strategies and handle the “no-retrieval” case, we introduce a baseline score threshold θ_{base} . This threshold represents the utility of providing no document at all. A document d_i is kept only if its score surpasses this baseline:

$$\mathcal{D}_{\text{filt}} = \{d_i | d_i \in \mathcal{D}_{\text{ret}} \wedge \mathcal{M}_{\text{score}}(x, d_i) > \theta_{\text{base}}\} \quad (9)$$

This allows the scorer model to discard any number of documents, including all of them if they are all predicted to be less useful than generating an answer with no external knowledge.

D. Inference Pipeline

We now outline the complete inference pipeline that integrates our knowledge filter into the RAG workflow, as illustrated in Fig. 2. Given a user query x , our pipeline first employs the retriever $\mathcal{R}$ to fetch a set of candidate documents $\mathcal{D}_{\text{ret}}$. This set is then passed to our trained Knowledge Filter $\mathcal{F}$ (instantiated as any of our four strategies), which evaluates each document and produces a refined subset $\mathcal{D}_{\text{filt}}$, by discarding any document deemed harmful. Finally, the contents of $\mathcal{D}_{\text{filt}}$ are concatenated with the original query and fed to the generator $\mathcal{M}$ to produce the final answer.

Crucially, our framework is designed to handle cases where all retrieved information is detrimental. If the filter discards every document, the resulting set $\mathcal{D}_{\text{filt}}$ becomes empty. In this scenario, the pipeline gracefully degenerates to a standard generation task, prompting the LLM with only the original query. This allows the model to respond using its internal parametric knowledge, ensuring it is shielded from harmful context while still addressing the user’s request.

IV. EXPERIMENT

To evaluate the effectiveness of our proposed framework and filtering strategies, we conduct a series of experiments on several benchmark question-answering datasets.

A. Experiment Setup

Datasets: We evaluate our proposed framework on three open-domain question answering datasets to assess its effectiveness across different types of questions and complexities:

- *Natural Questions (NQ)* [40]: A benchmark consisting of real user queries submitted to Google Search. It requires models to find long-form or short-form answers within a given Wikipedia article, testing real-world QA capabilities.
- *PopQA* [41]: A challenging dataset focusing on questions about popular entities where answers are often short and specific. It is designed to rigorously test the factual recall and accuracy of language models.
- *AmbigNQ* [42]: A dataset derived from NQ that specifically targets ambiguous questions. To be considered fully correct, a model must produce all possible valid answers, which tests its ability to handle ambiguity and provide comprehensive responses.

Due to computational constraints, we randomly sample 500 instances from the official test set of each dataset for our evaluation. If a dedicated test set is unavailable, we sample from the validation set.

Implementation Details: Our experimental pipeline is built upon the following components and configurations. We test our framework using two powerful open-source instruction-tuned models as the backbone generator: LLaMA2-7B-Chat¹ [33] and Qwen2.5-7B-Instruct² [34]. For retrieval, we utilize the BM25 algorithm. The external knowledge corpus is the English Wikipedia dump [43] from August 1, 2019. For each query, our RAG process begins by retrieving the Top-100 documents with BM25. This initial set is then processed by our knowledge filtering module. From the documents that are not filtered out, we select the Top-3 to be concatenated with the original query and fed to the generator. Regarding our trainable filtering strategies, the Proxy Model is built upon a deberta-v3-base³ [38] backbone, while the Preference-Aligned Scorer uses xlm-roberta-base⁴ [39] as its backbone. During the training of these filter models, we perform a hyperparameter search for the learning rate over the set {1e-6, 5e-6, 1e-5, 5e-5, 1e-4} and use a fixed batch size of 4. All experiments are conducted on a single NVIDIA Tesla A100 GPU with 40 GB of memory. We evaluate all methods using two standard metrics for open-domain question answering: Hit@1, which measures whether the generated answer exactly matches any of the ground-truth answers, and the F1 Score, which measures the token-level overlap between the prediction and a ground-truth answer. To ensure rigorous evaluation, we constructed balanced training datasets for each

¹<https://huggingface.co/meta-llama/Llama-2-7b-chat-hf>

²<https://huggingface.co/Qwen/Qwen2.5-7B-Instruct>

³<https://huggingface.co/microsoft/deberta-v3-base>

⁴<https://huggingface.co/FacebookAI/xlm-roberta-base>

TABLE I

MAIN EXPERIMENTAL RESULTS COMPARING OUR PROPOSED METHODS AGAINST BASELINES ON THREE QA DATASETS. WE REPORT HIT@1 AND F1 SCORES. THE BEST PERFORMANCE IN EACH COLUMN IS HIGHLIGHTED IN **BOLD**.

Method	NQ		PopQA		AmbigNQ		Average	
	Hit@1	F1	Hit@1	F1	Hit@1	F1	Hit@1	F1
<i>LLaMA2-7B-Chat</i>								
Vanilla Chat	33.20	38.48	20.00	23.52	37.00	41.71	30.07	34.57
Direct RAG	35.00	39.86	27.20	29.81	43.00	47.18	35.07	38.95
<i>Baselines</i>								
RECOMP	33.80	37.55	26.40	28.13	41.60	45.80	33.93	37.16
CompAct	34.20	38.10	26.80	28.90	42.00	46.24	34.33	37.75
EXIT	36.40	41.33	29.00	31.65	44.20	49.02	36.53	40.67
Self-Ask	35.80	40.71	28.20	30.25	45.00	50.11	36.33	40.36
SKR-KNN	37.20	42.15	31.00	33.54	45.80	50.80	38.00	42.16
SlimPLM	39.00	43.88	33.80	36.91	47.20	51.75	40.00	44.18
Ensemble (ACL'25)	39.50	44.20	34.10	37.15	47.50	51.60	40.37	44.32
Ours	40.20	45.15	35.00	38.12	48.40	52.68	41.20	45.32
<i>Qwen2.5-7B-Instruct</i>								
Vanilla Chat	26.40	31.86	17.60	20.61	28.80	35.87	24.27	29.45
Direct RAG	36.00	41.00	27.80	30.92	38.80	45.22	34.20	39.05
<i>Baselines</i>								
RECOMP	34.60	38.75	26.80	29.50	37.20	43.14	32.87	37.13
CompAct	35.20	39.40	27.00	30.15	38.00	44.25	33.40	37.93
EXIT	37.20	42.85	29.20	32.78	40.60	47.90	35.67	41.18
Self-Ask	36.80	42.10	28.60	31.66	41.00	48.21	35.47	40.66
SKR-KNN	38.00	43.55	30.20	34.12	42.00	49.30	36.73	42.32
SlimPLM	39.40	45.02	31.20	35.45	43.20	50.81	37.93	43.76
Ensemble (ACL'25)	40.00	45.10	31.70	35.70	43.10	50.80	38.27	43.87
Ours	40.80	46.25	32.40	36.88	44.00	51.92	39.07	45.02

task. Specifically, we utilized 5,000 samples for the Contrastive Pre-training stage and 10,000 samples for the Direct Preference Optimization stage. For methods requiring Supervised Fine-Tuning, such as the Proxy Model, we also employed 10,000 samples. In all sets, the ratio of 'Harmful' to 'Harmless' labels is strictly maintained at 1:1 to avoid class imbalance.

B. Main Result

We compare our proposed framework against a comprehensive set of baselines, which can be divided into three main categories: *fundamental approaches*, *pre-retrieval methods*, and *post-retrieval methods*.

Fundamental approaches: This category establishes the performance of standard, unmodified pipelines, serving as essential reference points for comparison.

- *Vanilla Chat*: This is the most basic baseline, where the language model answers the query directly using only its internal parametric knowledge, without any retrieval augmentation.
- *Direct RAG*: This represents the standard, non-adaptive RAG pipeline. It retrieves the Top-k documents using BM25 and directly concatenates their content with the query to feed into the generator.

Pre-retrieval methods: These methods aim to determine if or what information is needed before or during the retrieval process.

- *Self-Ask [44]*: This method prompts the language model to decompose a complex query into a series of simpler, follow-up sub-questions. These sub-questions are then answered, often with the help of a search engine, to construct the final answer.

- *SKR-KNN [9]*: This approach decides whether retrieval is necessary by first finding the k-nearest neighbor questions from the training set. The final decision is based on the majority vote of whether these neighboring questions required retrieval.
- *SlimPLM [22]*: This framework uses a much smaller, lightweight model to predict what knowledge the main LLM is missing for a given query. Retrieval is then performed only for this predicted missing knowledge, aiming to reduce the inference cost of the LLM.

Post-retrieval methods: These methods focus on condensing the retrieved documents before they are passed to the generator.

- *RECOMP [45]*: This framework compresses retrieved documents into concise textual summaries. It can operate either extractively (selecting key sentences) or abstractively (generating a new summary) to reduce context length and computational cost.
- *CompAct [46]*: This method employs an active, iterative strategy to condense extensive documents. It is designed as a plug-in module to filter out irrelevant information and achieve high compression rates.
- *EXIT [12]*: An extractive compression framework that performs parallelizable, context-aware sentence selection. It aims to improve both accuracy and efficiency by adapting to query complexity and retrieval quality.
- *Ensemble (ACL'25) [1]*: This baseline adapts the method from our previous work, originally proposed for discriminative tasks.

As shown in Table I, we present the experimental results of our proposed method against all baselines on three QA datasets. As expected, the Vanilla Chat model, which relies solely on

TABLE II
ABLATION STUDY COMPARING THE PERFORMANCE OF OUR FOUR PROPOSED KNOWLEDGE FILTERING STRATEGIES

Method	NQ		PopQA		AmbigNQ		Average	
	Hit@1	F1	Hit@1	F1	Hit@1	F1	Hit@1	F1
<i>LLaMA2-7B-Chat</i>								
Vanilla Chat	33.20	38.48	20.00	23.52	37.00	41.71	30.07	34.57
Direct RAG	35.00	39.86	27.20	29.81	43.00	47.18	35.07	38.95
<i>w/ Knowledge Filtering</i>								
Direct Prediction	36.20	40.63	28.00	30.53	40.40	44.75	34.87	38.64
In-Context Prediction	37.00	41.27	27.20	29.82	43.80	48.12	36.00	39.74
Proxy Model	38.80	43.28	33.20	36.44	46.60	50.94	39.53	43.55
Preference-Aligned Scorer	40.20	45.15	35.00	38.12	48.40	52.68	41.20	45.32
<i>Qwen2.5-7B-Instruct</i>								
Vanilla Chat	26.40	31.86	17.60	20.61	28.80	35.87	24.27	29.45
Direct RAG	36.00	41.00	27.80	30.92	38.80	45.22	34.20	39.05
<i>w/ Knowledge Filtering</i>								
Direct Prediction	38.00	43.08	28.80	32.22	40.00	46.37	35.60	40.56
In-Context Prediction	38.40	43.60	27.80	31.43	41.20	47.96	35.80	41.00
Proxy Model	39.20	44.15	31.00	34.63	42.20	49.66	37.47	42.81
Preference-Aligned Scorer	40.80	46.25	32.40	36.88	44.00	51.92	39.07	45.02

its parametric knowledge, consistently yields the lowest performance across all datasets, underscoring the necessity of retrieval augmentation for these knowledge-intensive tasks. The Direct RAG approach establishes a solid baseline, significantly improving upon Vanilla Chat by incorporating retrieved documents. This validates the efficacy of incorporating external knowledge to enhance generation quality for QA tasks.

Among the more advanced approaches, the context compression methods demonstrate variable efficacy. Strategies such as RECOMP and CompAct yield only marginal gains over Direct RAG, and in some cases underperform, indicating that indiscriminate compression risks the omission of relevant information. In contrast, adaptive retrieval methods represent a more effective paradigm. By making more discerning decisions about when or what to retrieve, SlimPLM consistently achieves the highest performance among all other baselines. This illustrates the superiority of strategies that mitigate the risk of introducing distracting noise from unnecessary retrieval over static context augmentation.

Most notably, our proposed framework demonstrates superior performance, consistently outperforming all baseline methods across both backbone models and all three datasets. These results strongly indicate that our fine-grained filtering mechanism is a more potent strategy for enhancing RAG performance. By learning to explicitly identify and remove documents harmful to the final generation, our approach more effectively refines the context provided to the LLM compared to both context compression and pre-retrieval gating techniques.

C. Impact of Filtering Strategies

We evaluate the effectiveness of our four proposed knowledge filtering strategies with the results presented in Table II. This analysis allows us to understand the contribution of each strategy's design choices, from simple prompting to sophisticated preference alignment.

As shown in the table, all four filtering methods consistently improve upon the Direct RAG baseline, confirming the overall advantage of actively filtering retrieved knowledge. The most basic approach, Direct Prediction, which relies on the LLM's zero-shot judgment, offers a modest improvement. This suggests that while the LLM has some inherent ability to identify unhelpful context, its general-purpose reasoning is not perfectly aligned with our performance-oriented definition of "Harmful." Providing several task demonstrations in the In-Context Prediction strategy yields a further performance gain, demonstrating that guiding the LLM with task-specific demonstrations helps it better grasp the desired filtering behavior. A significant leap in performance is observed with the Proxy Model. By training a dedicated classifier on the entire labeled dataset, this strategy substantially outperforms the two LLM-prompting methods. This highlights the limitations of in-context learning and validates the effectiveness of fine-tuning a specialized model that can learn from a much larger set of examples. Finally, the Preference-Aligned Scorer achieves the best performance across all datasets and backbone models. This superiority underscores the advantages of our two-stage training process. By first using contrastive learning to structure the representation space and then applying Direct Preference Optimization, the scorer learns a more effective scoring function than the simple binary classification of the Proxy Model. This progression of results strongly validates our design, showing that framing the problem as preference alignment is the most effective way to learn and apply the desired filtering logic.

V. ANALYSIS AND DISCUSSION

A. Performance of Strategies on Knowledge Filtering

To directly assess the capability of our four proposed strategies to identify harmful documents, we evaluate their classification performance on the test sets. We report the Precision, Recall, and F1-score for the "Harmful" class, with the results detailed in Table III.

TABLE III
FILTERING PERFORMANCE OF THE FOUR PROPOSED STRATEGIES IN IDENTIFYING HARMFUL DOCUMENTS, USING THE LLAMA2-7B-CHAT BACKBONE. WE REPORT PRECISION (P), RECALL (R), AND F1-SCORE (F) FOR THE “HARMFUL” CLASS.

Method	NQ			PopQA			AmbigNQ		
	P	R	F1	P	R	F1	P	R	F1
Direct Prediction	48.5	35.2	40.8	45.1	33.8	38.6	44.2	31.5	36.8
In-Context Prediction	68.2	96.5	79.9	65.4	100.0	79.1	63.9	94.1	76.2
Proxy Model	85.1	90.0	87.5	83.5	89.0	86.2	81.2	87.0	84.0
Preference-Aligned Scorer	90.5	94.0	92.2	89.5	93.0	91.2	87.5	91.0	89.2

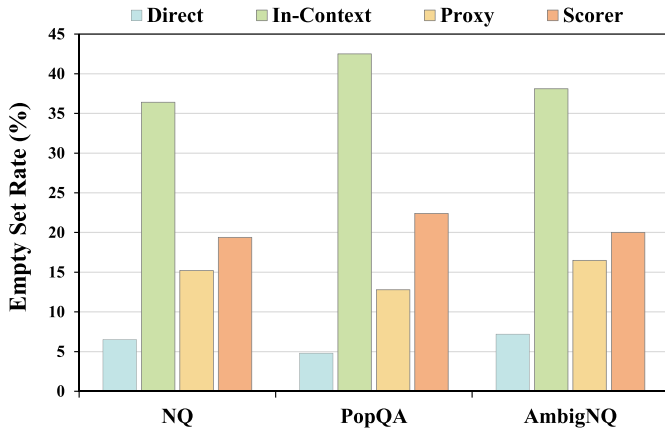


Fig. 5. Empty Set Rate of our four filtering strategies on the NQ, PopQA, and AmbigNQ datasets, using the LLAMA2-7B-Chat backbone. This rate represents the percentage of queries for which a given strategy discarded all retrieved documents, leading to a generation based solely on the model’s parametric knowledge.

The result reveals that Direct Prediction is a weak baseline, with low Recall indicating it fails to identify most harmful documents. The In-Context Prediction strategy achieves a substantial boost in Recall, even reaching 100% on the PopQA dataset by incorporating task-specific demonstrations. However, this high Recall comes at the expense of Precision. This indicates an excessive number of false positives, where the model becomes overly cautious and incorrectly classifies many harmless documents as harmful. By fine-tuning a dedicated classifier, the Proxy Model achieves a strong and balanced strategy of both Precision and Recall. This establishes a robust baseline for a trained filter and confirms the superiority of a dedicated model over prompting-based methods. Crucially, the Preference-Aligned Scorer demonstrates a clear advantage over all other methods, including the Proxy Model. This substantial performance gap highlights the power of our two-stage training process. The combination of contrastive learning to structure the embedding space and Direct Preference Optimization to fine-tune on preference pairs allows the scorer to learn a far more accurate decision boundary than is possible with standard binary classification.

B. Effects of Different Strategies on Empty Set Rate

To better understand the filtering behavior of each strategy, we measured the “empty set rate”—the frequency with which a filter discards all retrieved documents, thereby degenerating to a standard QA generation. The results, shown in Fig. 5, reveal the distinct characteristics of each method.

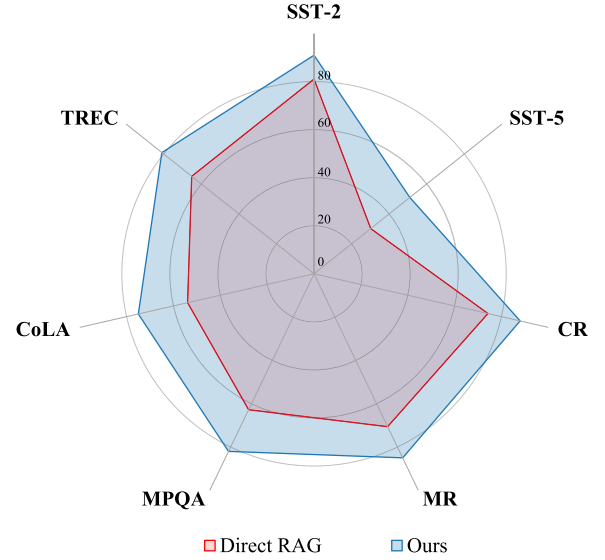


Fig. 6. Performance comparison of our proposed method (Ours) and the Direct RAG baseline on seven Non-Knowledge-Intensive datasets. The metric is Accuracy (%). The larger area covered by our method (blue) indicates superior performance across all tasks.

The Direct Prediction strategy exhibits a very low empty set rate (below 8% on all datasets), indicating a conservative nature where it rarely deems an entire set of documents to be harmful. In contrast, In-Context Prediction shows an exceptionally high empty set rate, exceeding 35% on all datasets. This aggressive behavior, coupled with its lower filtering F1-score (Table III), suggests that it is prone to false positives, often incorrectly discarding all context and forcing a fallback to the weaker Vanilla Chat performance. The two trained models, the Proxy Model and Preference-Aligned Scorer, demonstrate a more balanced and moderate empty set rate. Notably, the scorer consistently has a slightly higher empty set rate than the Proxy Model across all three datasets. This observation, when considered alongside the superior downstream performance of the scorer (as shown in Table II), is particularly insightful. It suggests that the scorer is not just more accurate at identifying individual harmful documents but is also more adept at recognizing challenging cases where the entire retrieved set is detrimental.

C. Performance of Filtering Strategies on Non-Knowledge-Intensive Tasks

To evaluate the generalizability and robustness of our framework beyond its primary application in QA, we extend its evaluation to a suite of Non-Knowledge-Intensive (NKI) text

TABLE IV

CASE STUDY ON THE QUERY “WHO IS THE CAST OF THE NEW TOMB RAIDER MOVIE”. THE PREFERENCE-ALIGNED SCORER SUCCESSFULLY DISTINGUISHES BETWEEN DOCUMENTS REFERENCING THE 2001 FILM (HARMFUL) AND THE 2018 REBOOT (HARMLESS), ENABLING THE GENERATOR TO PRODUCE THE CORRECT ANSWER. RED TEXT INDICATES REASONS FOR HARMFULNESS, GREEN TEXT INDICATES REASONS FOR UTILITY, AND BLUE TEXT REPRESENTS MODEL BEHAVIOR ANALYSIS.

Category	Content
Query	who is the cast of the new tomb raider movie?
Ground Truth	Alicia Vikander as Lara Croft, Dominic West , Walton Goggins , Daniel Wu , and Kristin Scott Thomas
Harmful Documents	Doc 1: “Lara Croft: Tomb Raider” ... featuring the character Lara Croft, portrayed by Angelina Jolie... <i>(Refers to 2001 film)</i> Doc 2: ...whoever Hollywood producers cast in the planned reboot... <i>(Speculation/Irrelevant)</i> Doc 3: ...elevates goofiness to an art form... <i>(Review of 2001 film)</i>
Harmless Documents	Doc 1: ...Showtime paid \$6.8 million... The deal between Eidos... <i>(Business context, non-contradictory)</i> Doc 2: Tomb Raider... is a media franchise... <i>(General Definition)</i> Doc 3: Tomb Raider (film)... is a 2018 action-adventure film directed by Roar Uthaug... <i>(Correct 2018 Context)</i>
Vanilla Chat (F1: 0.0000)	Angelina Jolie <i>(Error Analysis: Hallucination based on internal knowledge of the older movie)</i>
Direct RAG (F1: 0.0455)	The context provided does not list the cast of the new Tomb Raider movie. It discusses... <i>(Error Analysis: Confused by conflicting dates and reviews, refuses to answer)</i>
Ours (F1: 0.4516)	The main cast of the new Tomb Raider movie (2018) includes Alicia Vikander as Lara Croft, Dominic West, Walton Goggins, and Daniel Wu. <i>(Success: Correctly identifies the 2018 cast by filtering out the 2001 context)</i>

classification tasks. We selected seven widely-used datasets: SST-2 [47], SST-5 [47], CR [48], MR [49], MPQA [50], CoLA [51] and TREC [52]. For this experiment, we compare our best-performing strategy, the Preference-Aligned Scorer, against the Direct RAG baseline. We use BM25 for retrieval and report the classification Accuracy (ACC) as the evaluation metric.

As shown in Fig. 6, the result reveals a clear and consistent trend: our proposed method (Ours) significantly outperforms the Direct RAG baseline across all seven text classification datasets. This finding is particularly insightful, as it indicates that the utility of our filtering mechanism is not confined to fact-based tasks. On these NKI tasks, “Harmful” knowledge is not necessarily factually incorrect but rather contextually distracting or irrelevant. For instance, in a sentiment analysis task, a retrieved document might contain conflicting sentiment signals or be semantically unrelated, leading the standard RAG model to make an incorrect classification. Our method demonstrates its effectiveness in identifying and eliminating this harmful context, allowing the LLM to make a more accurate judgment based on the input text.

D. Case Study

To provide deeper insight into how our Knowledge Filtering framework improves generation reliability, we conduct a qualitative analysis using a representative example from the test set, as shown in Table IV. The user query, “who is the cast of the new tomb raider movie?” presents a typical challenge of temporal ambiguity and entity disambiguation. The term “new” implies the user is interested in the 2018 reboot, but the retriever (BM25) fetches a mix of documents related to both the 2018 film and the original 2001 adaptation starring Angelina Jolie.

In the standard Vanilla Chat setting, the model fails to capture the “new” constraint and hallucinates the cast of the older movie (“Angelina Jolie”), resulting in a zero F1 score. Similarly, the Direct RAG approach is overwhelmed by the contradictory noise. The retrieved context contains reviews and plot descriptions of the 2001 film (labeled as Harmful Documents in the table), which conflict with the information about the 2018 version. Unable to reconcile these conflicts, the generator essentially gives up, stating that the context does not list the cast.

In contrast, our Preference-Aligned Scorer successfully assigns lower scores to the documents discussing the 2001 film and irrelevant commentary, effectively filtering them out. Crucially, it retains the document describing the “2018 action-adventure film” (Harmless Doc 3) and other non-contradictory business context. By removing the misleading information about the previous adaptation, our method clarifies the input context. This allows the LLM to correctly identify the 2018 cast members, such as Alicia Vikander and Dominic West, boosting the F1 score to 0.45. This case demonstrates that “harmful knowledge” is often contextually misleading rather than factually wrong in isolation, and our method effectively purges such distractors to align the retrieval context with the user’s specific intent.

E. Computational Efficiency Analysis

To further validate the practicality of our proposed framework, we conducted a quantitative evaluation of the computational overhead introduced by the knowledge filtering module. We compared our approach against three representative paradigms: Standard RAG, which directly feeds retrieved documents to the generator; SlimPLM, which employs the LLM itself to self-assess the need for retrieval (often requiring a two-step

TABLE V

COMPARISON OF COMPUTATIONAL EFFICIENCY ACROSS DIFFERENT RAG PARADIGMS. WE REPORT THE AVERAGE LATENCY (MS), P95 LATENCY (MS), THROUGHPUT (QUERIES/SECOND), AND PEAK GPU MEMORY (MB). “OURS” REPRESENTS THE TOTAL COST OF THE STANDARD RAG GENERATOR PLUS OUR KNOWLEDGE FILTERING MODULE.

Method	Avg. Latency (ms)	P95 Latency (ms)	Throughput (q/s)	Peak Memory (MB)
Standard RAG (No Filtering)	818.56	825.03	1.22	3702.88
SlimPLM	1580.40	1610.15	0.63	4950.20
EXIT	1350.12	1395.80	0.74	4680.50
Ours (RAG + Filtering)	1139.19	1147.15	0.88	4406.40

inference process); and EXIT, which utilizes an additional model to summarize context before generation. All experiments were conducted on a single NVIDIA A100 GPU, reporting the average latency per query and peak GPU memory usage.

As presented in Table V, the Standard RAG pipeline naturally exhibits the lowest latency and memory footprint as it involves a single pass of the generator. However, this comes at the cost of potential hallucinations caused by unfiltered noise. The Retrieval Necessity and Document Compression methods, while attempting to mitigate noise, incur substantial computational costs. The Necessity approach nearly doubles the latency due to the requirement of self-reflection passes by the large language model, while Compression techniques demand significant memory resources to handle the summarization of multiple documents. In contrast, our approach demonstrates a superior trade-off between performance and efficiency. By offloading the filtering task to a lightweight Preference-Aligned Scorer (based on XLM-Roberta), we introduce only a marginal latency overhead of approximately 320 ms compared to the Standard RAG baseline. Furthermore, the memory consumption of our method remains significantly lower than that of the competitive baselines, as our scorer is considerably smaller than the generative backbone or complex compression models. This analysis confirms that our framework effectively filters harmful knowledge without imposing a prohibitive computational burden, making it highly suitable for real-time applications.

F. Human Evaluation and Limitation

To rigorously assess the validity of our proposed Annotation strategy and ensure its alignment with genuine document utility, we conducted a human evaluation study. Specifically, we randomly sampled 100 query-document pairs from our annotated training set, ensuring a balanced distribution of samples labeled as “Harmful” and “Harmless” by our pipeline. Two independent experts with backgrounds in NLP were then tasked with manually evaluating these pairs based on a strict criterion: whether the document contained information that factually contradicted the ground truth or introduced irrelevant noise likely to mislead a reasoner. The empirical results demonstrate a high degree of consistency between our automated strategy and human judgment, achieving an agreement rate of 88% and a Cohen’s Kappa coefficient of 0.76.

A closer examination of the discrepancies (approximately 12% of cases) reveals that most disagreements arose in

borderline scenarios. In these instances, documents often appeared semantically relevant to human annotators but contained subtle temporal ambiguities or factual nuances that led the model to hallucinate. Our automated strategy correctly identified these as “Harmful” due to the observed performance degradation, suggesting that the approach effectively captures specific model-centric utility that might occasionally elude manual inspection.

Notwithstanding these positive findings, we acknowledge that the Annotation strategy has inherent limitations. Since the harmfulness labels are derived directly from variations in the generator’s performance, the quality of the resulting dataset is intrinsically tied to the reasoning capabilities of the backbone LLM used during the annotation phase. If the generator exhibits unstable reasoning or extreme hallucination independent of the retrieval context, it may introduce noise into the training data. Future work may explore cross-model annotation techniques to mitigate this dependency on a specific model’s inductive biases.

VI. CONCLUSION

In this paper, we addressed the critical challenge of harmful knowledge within RAG systems, where retrieved documents can be irrelevant, misleading, or factually incorrect, thereby degrading the quality of the final generated output. We proposed a novel, two-stage framework designed to first identify and then filter out documents that degrade the quality of generated answers. Our approach begins with a Label Collection strategy, where we define the harmfulness of a document based on its impact on the downstream task performance of the Large Language Models. This task-oriented definition provides a solid foundation for training our second module. The annotated data provide the crucial supervisory signal required to train our Knowledge Filtering module. Within this module, we introduced and thoroughly evaluated four distinct filtering strategies, leveraging techniques from LLM self-evaluation to preference alignment. Extensive experiments conducted on three diverse Question Answering datasets validate the effectiveness of our framework. The results demonstrate that our approach can reliably identify and eliminate harmful knowledge from the retrieved documents, leading to significant improvements in the accuracy and reliability of the final generated answers.

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